Cameron Harvey, cnh43, MAE 324, HW #5, 3/10/26

#1 Known: Cylindrical fin $r = 1mm$, $L = 5cm$, $T_\infty = 25°C$, $h = 10 \frac{W}{m^2K}$, $k = 200 \frac{W}{mK}$

Tip of fin is thermally insulated

Find: a) If $T_b = 125°C$ find $T = L$ & $\frac{L}{2}$

b) ε_c & η_c using 3.4

c) η_f using 3.5 & discuss

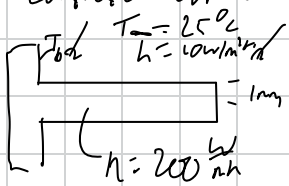
d) R_f

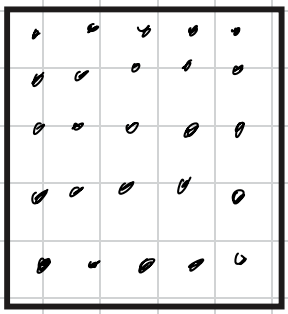
e) Fins form 5x5 array on plate $s = 3cm$ & calc η_o & R_o & $q_{unfinned}$ & q_{fin}

f) Array calc as heat sink mounted onto slab with $L = 2cm$, $k_1 = 10 W/mK$ & adhesive $R_c = 10^{-4} m^2K/W$, back of heater is insulated, $q'' = 1000 W/m^2$

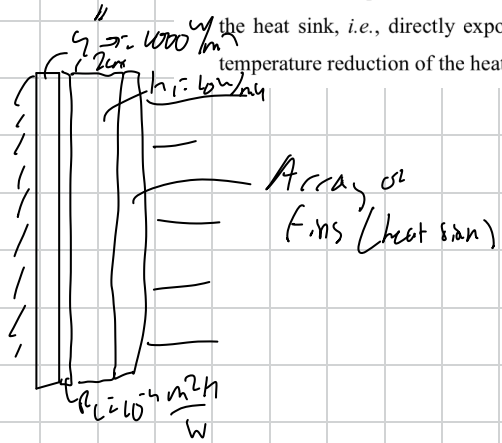
Draw thermal resistance network of this system & calc. temp of Thin film heater T_h & compare without heatsink

Schematic:

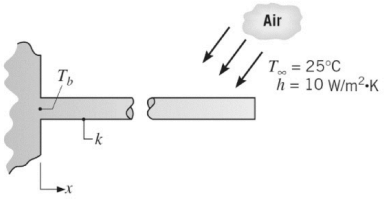




3cm



1. (60 pt) A cylindrical pin fin with a radius of $r = 1$ mm and length of $L = 5$ cm is exposed to airflow with a temperature of $T_\infty = 25$ °C and a heat transfer coefficient of $h = 10$ W/m²·K. Thermal conductivity of the pin fin is $k = 200$ W/m·K. The tip of the pin fin is thermally insulated. Use the fin problem solving workflow discussed in the lecture to solve the following questions.



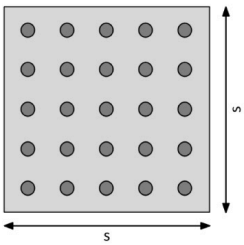
(a) If the fin base temperature is $T_b = 125$ °C, please calculate the fin temperature at $x = L/2$ and $x = L$.

(b) Calculate the fin effectiveness ε_f and the fin efficiency η_f based on their definition and results provided by Table 3.4 of the textbook.

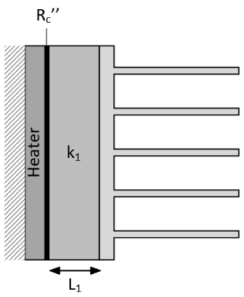
(c) Calculate the fin efficiency η_f based on results provided by Table 3.5. Compare the results of (b) and (c) and discuss if Table 3.5 provided a good approximation to the fin efficiency.

(d) Calculate the thermal resistance of this pin fin R_f.

(e) The pin fins then form a 5×5 array mounted on a base plate with a side length of $s = 3$ cm, as shown in the figure on the right-hand side. Calculate the overall efficiency of the fin array η_o and the thermal resistance of the fin array R_o. Calculate how much heat is dissipated through the unfined surface and how much is through the fins.



(f) The heat sink consisting of this fin array is used to improve the heat dissipation from a thin film heater, as shown in the figure on the right-hand side. The heat sink is mounted on a slab with thickness $L_1 = 2$ cm and thermal conductivity $k_1 = 10$ W/m·K. There is a thermal adhesive between the thin film heater and the slab, resulting in a contact thermal resistance $R_c = 10^{-4}$ m²·K/W. Thermal insulation was applied to the backside of the heater. The heater generates a uniform heat flux $q'' = 1000$ W/m², which was dissipated through the heat sink. The temperature across the thin film heater is uniform. Thermal resistance for the base of the heat sink can be neglected. There is no contact thermal resistance between the base of the heat sink and the slab. Please draw the thermal resistance network of this system and calculate the temperature of the thin film heater T_h. Compared with the condition without the heat sink, *i.e.*, directly exposing the heater and the slab to ambient air, how much temperature reduction of the heater is achieved?



Assumptions: Steady State, 1-D heat transfer through fin, Uniform material properties, Perfectly insulated, Fin assumption ↑.

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#1 cont.

Analysis:

a) $\theta = T - T_\infty$ $\qquad \frac{\theta}{\theta_b} = \frac{\cos\sqrt{\frac{hP}{kA_c}}(L-x)}{\cosh(\sqrt{\frac{hP}{kA_c}}\cdot L)}$

$\theta_b = T_b - T_\infty$

$$\theta(b) = \frac{\cos\sqrt{\frac{hP}{kA_c}}(L-x)}{\cosh(\sqrt{\frac{hP}{kA_c}}\cdot L)}\cdot(T_b - T_\infty)$$

$h = 10\frac{W}{m^2K}$ $\qquad T_b = 125°C$

$P = 2\pi\cdot 0.001m$ $\qquad T_\infty = 25°C$

$A_c = \pi\cdot(0.001m)^2$ $\qquad L = 5cm$

$k = 200\frac{W}{mK}$

$$\theta(b) = \frac{\cosh(10(0.05-x))}{1.127}\cdot 100°C$$

$$\theta(x=L) = \frac{1}{1.127}\cdot 100°C = 88.68°C$$

$T_{b=L} = 88.68°C + 25°C$

$T(x=L) = 113.68°C$

$$\theta(b=\tfrac{L}{2}) = \frac{1.03}{1.127}\cdot 100 = 91.47°C$$

$T_{x=L/2} = 91.47°C + 25°C$

$T(x=L/2) = 116.47°C$

b) $\varepsilon_f = \frac{q_f}{q_{nf}}$ $\qquad \eta_f = \frac{q_f}{q_{max}}$

$$q_f = \frac{\sqrt{hPkA_c}\,\theta_b \tanh(\sqrt{\frac{hP}{kA_c}}\cdot L)}{hA_c\theta_b} = 92.42 = \varepsilon_f$$

$$\eta_f = \frac{\sqrt{hPkA_c}\,\theta_b \tanh(\sqrt{\frac{hP}{kA_c}}\cdot L)}{h\cdot A_f\cdot\theta_b} = 0.924 = \eta_f = 92.4\%$$

$(2\pi r\cdot L)$

c) $$\eta_f = \frac{\tanh(\sqrt{\frac{hP}{kA_c}}(L+(P/4)))}{\sqrt{\frac{hP}{kA_c}}(L+(P/4))}$$

$\eta_f = 0.923 = 92.3\%$

Compared to part b, we are a little low, only a 0.1% difference. So I would say the approximation is quite good

d) $$R_f = \frac{\theta_b}{q_f} = \frac{100°C}{0.290} = 344.4\frac{K}{W} = R_f$$

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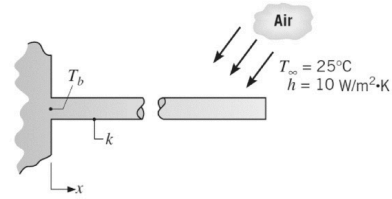

(a) If the fin base temperature is $T_b = 125$ °C, please calculate the fin temperature at $x = L/2$ and $x = L$.

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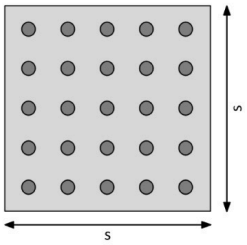
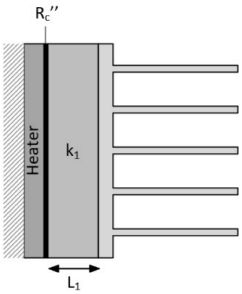


(f) The heat sink consisting of this fin array is used to improve the heat dissipation from a thin film heater, as shown in the figure on the right-hand side. The heat sink is mounted on a slab with thickness $L_1 = 2$ cm and thermal conductivity $k_1 = 10$ W/m·K. There is a thermal adhesive between the thin film heater and the slab, resulting in a contact thermal resistance $R_c = 10^{-4}$ m²·K/W. Thermal insulation was applied to the backside of the heater. The heater generates a uniform heat flux $q'' = 1000$ W/m², which was dissipated through the heat sink. The temperature across the thin film heater is uniform. Thermal resistance for the base of the heat sink can be neglected. There is no contact thermal resistance between the base of the heat sink and the slab. Please draw the thermal resistance network of this system and calculate the temperature of the thin film heater T_h. Compared with the condition without the heat sink, *i.e.*, directly exposing the heater and the slab to ambient air, how much temperature reduction of the heater is achieved?



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#1 cont.

Analysis cont.

eq $\eta_0 = 1 - \frac{N A_f}{A_{tot}}(1-\eta_f)$

$N = 25 \quad A_f = (2\pi \cdot 0.001m \cdot 0.05m)$

$A_{tot} = 25 \cdot A_f + (5.5 - 25 \cdot \pi \cdot .001m^2)$

$A_f = 3.1415 \cdot 10^{-4} m^2$

$A_{tot} = 0.00868 m^2$

$\eta_f = 92.4\%$

$\eta_0 = 1 - \frac{25 \cdot 3.1415 \cdot 10^{-4} m^2}{0.00868 m^2}(1 - .924)$

$\boxed{\eta_0 = 0.931 = 93.1\%}$

$R_0 = \frac{1}{\eta_0 h A_{tot}} = \frac{1}{.931 \cdot 10 \frac{W}{m^2 K} \cdot 0.00868 m^2}$

$\boxed{R_0 = 12.38 \ K/W}$

$q_{un} = h \cdot A_{un} \cdot \theta_b$

$= 10 \frac{W}{m^2 K} \cdot (5.5 - 25 \cdot \pi \cdot .001 m^2) \cdot 100°C$

$\boxed{q_{un} = 0.821 W}$

$q_{tot} = \frac{\theta_b}{R_0} = \frac{100°C}{12.38 K/W} = 8.08 W = q_{tot}$

$q_{fins} = q_{tot} - q_{un} = 8.08 W - 0.821 W$

$\boxed{q_{fins} = 7.26 W}$

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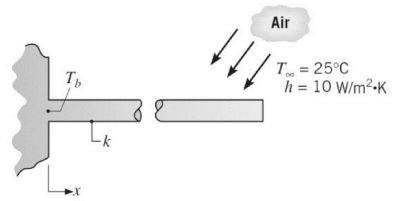

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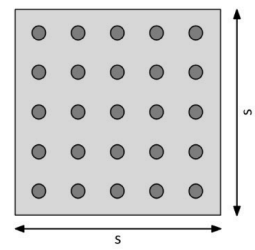

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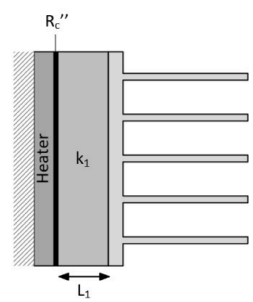

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#1 cont.
Analysis cont.

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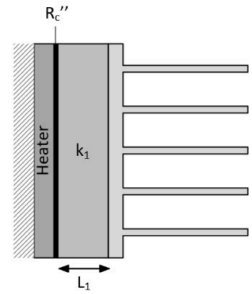
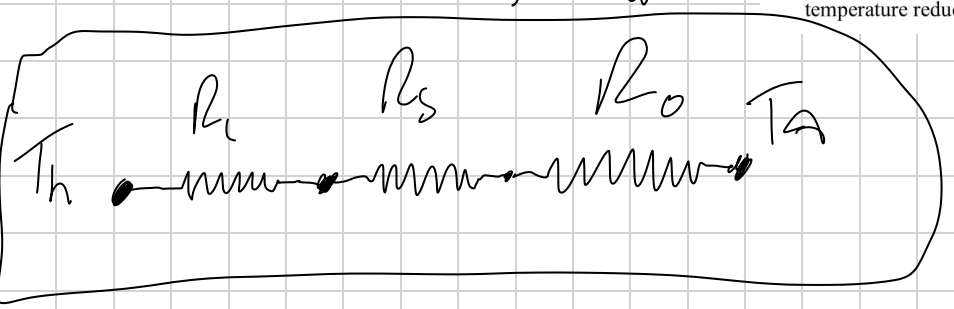


f) $R_c = R_c''/A = \frac{10^{-4}\ m^2K/W}{(.03m)^2}$

$R_c = 0.111\ \frac{K}{W}$

$R_s = \frac{L}{kA} = \frac{0.02m}{10\frac{W}{mK}\cdot(.03m)^2}$

$R_s = 2.22\ \frac{K}{W}$



$R_{eq} = R_c + R_s + R_o$

$= 0.111\ \frac{K}{W} + 2.22\ \frac{K}{W} + 12.38\ \frac{K}{W}$

$R_{eq} = 14.71\ K/W$

$q = 1000\ W/m^2 \cdot (0.03m)^2 = 0.9\ W = q$

$\Delta T = q \cdot R_{eq}$

$\Delta T = 0.9 \cdot 14.71$

$\Delta T = 13.23°C$

$T_h = 13.23°C + T_A$

$T_h = 38.24°C$

$R_{conv} = \frac{1}{hA}$

$= \frac{1}{10\frac{W}{m^2K}\cdot 0.03^2}$

$R_{conv} = 111.1\ \frac{K}{W}$

$\Delta T = q(R_{conv} + R_s + R_c)$

$T_h = 127.1°C$

↳ if just exposed heater

$\Delta T_h = 127.1°C - 38.24°C$

$\Delta T_h = 88.86°C$ reduced by adding the heat sink